

IB CS

IA Guidance

The Internal Assessment (IA)

Solution
Duration: 30 hours

Weighting at SL: 30%
Weighting at HL: 20%

Introduction The requirement of the internal assessment is to develop a solution for a specified client to a specified problem or an unanswered question. The solution is assessed using five criteria.

- Planning
- Solution overview
- Development
- Functionality and extensibility of product
- Evaluation

The Internal Assessment (IA)

Time allocation

It is recommended that a total of approximately 30 teaching hours for both SL and HL should be allocated to the work.

This should include:

- time for the teacher to explain to students the requirements of the internal assessment, including codes of ethical behaviour and confidentiality
- class time for students to work on the internal assessment
- time spent by the student making arrangements to collect data as appropriate
- time for consultation between the teacher and each student
- time to review and monitor progress, and to check authenticity.

Additional time may be needed outside normal class time for students to work independently, such as acquiring additional skills required for the project and consulting with other people.

Syllabus component	Teaching hours	
	SL	HL
Core syllabus content		
SL/HL core The topics that must be studied, including some practical work, are: • Topic 1: System fundamentals (20 hours) • Topic 2: Computer organization (6 hours) • Topic 3: Networks (9 hours) • Topic 4: Computational thinking, problem-solving and programming (45 hours)	80	80
HL extension The topics that must be studied, including some practical work, are: • Topic 5: Abstract data structures (23 hours) • Topic 6: Resource management (8 hours) • Topic 7: Control (14 hours)	---	45
Case study Additional subject content introduced by the annually issued case study	---	30
Option		
SL/HL core	30	30
HL extension Students study one of the following options: Option A: Databases Option B: Modelling and simulation Option C: Web science Option D: Object-oriented programming (OOP)	---	15
Internal assessment		
Solution Practical application of skills through the development of a product and associated documentation	30	30
Group 4 project	10	10
Total teaching hours	**150**	**240**

Project

Choice of topic

In identifying a problem, students can select any topic that interests them. It does not have to be directly related to the specified themes in the syllabus or to the option studied.

Students should undertake a challenging task using appropriate techniques to showcase their algorithmic thinking and organizational skills.

The solution may take one of these forms:

- Creating a new system, such as an OOP program, a relational database, a simulation or a stand-alone/web-based application
- Adding functionality to an existing system, such as connecting a webpage(s) to a database, writing a function for Moodle, writing a plug-in, or developing a stand-alone application

It is essential that whatever form the solution takes it ensures the student can explicitly demonstrate and document his or her algorithmic thinking skills.

Process: 30 hours of class time (+30 hours outside of lessons)

Process

T1 - WEEK 3: REQUIREMENTS

Research the project (this can involve conversations your client, individually or in groups)
Define the requirements - what is the problem, what are you going to make, what are the success criteria
Draft your Criteria A response

T1 - WEEK 5: FUNCTIONAL DESIGN

Design the interface for your system and write a detailed functional description of how it will work.

T1 - WEEK 6: SYSTEM DESIGN

Think about the key functions or libraries you will use, and any objects you will create. Break the work to be done into separate chunks with limited interaction. Draw a structure chart or DFD.
Draft your Criteria B response

Process

T1 - WEEK 7: BEGIN DEVELOPMENT

What parts will you do first? Is there a particularly tricky/high risk bit you want to solve? Is there something you can get working by itself and then build on? Get some code up and running.

- Plan the rest of the project (and build a retrospective plan for what you've done so far) as a **Record of Tasks** (for Criteria B)
- **Create a Test Plan** (for Criteria B)

Write your Criteria C response as you go

T1 - WEEK 8 TO T2 WEEK 6 : DEVELOPMENT AND TESTING

Continue writing Criteria C as you go, and fill out your Test Plan when you get to that

T2 - WEEK 7-8: CLIENT REVIEW, DOCUMENTATION, SUBMISSION

Record your video for Criteria D, write Criteria E, submit by Monday of Week 9

Assessment

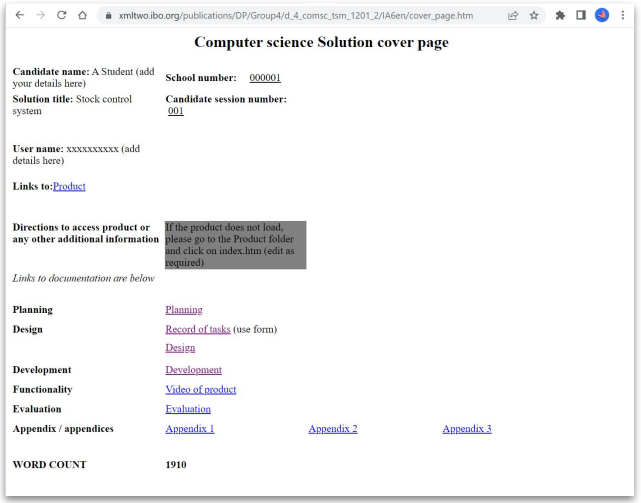
Submission

A .zip file



Requirements

1. A cover page
   - The cover page form to be used is included in HTML format within the zip file available on the OCC.
   - The cover page must be submitted in HTML format and should provide access to the product and associated documentation via relative hyperlinks.
   - The cover page is not included in the overall word count for the project, nor is its functionality assessed.
   - Information to access or locate the product, for example, a username and password, must be provided in the cell on the cover page.
   - The cover page must be called [cand_no]_[cand_name]_CoverPage.htm and be located in the top level folder.
2. The product
3. The documentation including a video (in a commonly used format such as .avi or .wmv)



The Product

Students should aim to develop a product that uses appropriate (complex) techniques, is fully functional and allows the moderator, if possible, access to its complete internal structure.

The moderator must be able to see the product functioning as a video. The video should address the success criteria stated in the planning criterion.

Any text within the product is not included in the overall word count for the solution.

Assessment Task:

Uploaded to Canvas

Before Period 1, Monday June 24

ZIP file, including ‘cover page’

To be assessed on Criteria A through E

Criteria for Internal Assessment

A: Planning (project scenario, success criteria)
B: Solution overview (record of tasks, design overview, test outline)
C: Development (narrative of tools and techniques)
D: Functionality and Extensibility of Product (video)
E: Evaluation (written assessment and recommendations)

Extensibility

In computing, ‘extensibility’ is the quality of being designed to allow the addition of new capabilities or functionality.

What factors might make a piece of software more or less extensible?

The table below indicates the content and nature of each of the files and the criterion that it relates to.

Document	Method of submission	Criterion
Description of scenario	Extended writing	A Planning
Rationale for the proposed product	Extended writing	A Planning
Success criteria for product	Bullet points	A Planning
Record of tasks	*Record of tasks* form	B Solution overview
Design overview	Design overview document, for example, screenshots, flowcharts, tables, diagrams	B Solution overview
Developing the product	Extended writing with screenshot evidence	C Development
The functioning product	Video (2–7 minutes in length) demonstrating the product	D Functionality and extensibility of product
Extensibility of product	Assessed through design overview and developing the product	D Functionality and extensibility of product
Evaluating the product	Extended writing	E Evaluation
Recommendations for improving the product	Extended writing	E Evaluation
Appendix	To contain additional information if appropriate	n/a

Prescribed format(s) of presentation are indicated in bold.

Criterion	Recommended word count	Comments—format of presentation
Planning		
Defining the problem or unanswered question	175–250	**Extended writing**
Justification for the proposed product	175–250	**Extended writing**
Success criteria	n/a	Bulleted list
Solution overview		
Record of tasks	n/a	***Record of tasks* form**: May be presented in the form of a table. (Note: **Use of template required**.)
Design overview	n/a	**Diagrams**: Including scanned sketches, lists or concise statements, table for assets and techniques and for test plan.
Development		
Developing the product	500–1,000	**Extended writing**: Justification of techniques used, including annotated screenshots and sources where appropriate.
Functionality and extensibility of product		
Evidence of functionality	n/a	**Video** of product functioning plus, if necessary or appropriate, document including screenshots showing the product functioning should be included in the documentation folder with information on how to access it on the cover page.
Extensibility	n/a	Assessed through design overview and, where appropriate, code listing.
Evaluation		
Evaluation of the product	175–250	**Extended writing**
Recommendations for further development	175–250	**Extended writing**
TOTAL	1,200–2,000	
Appendix/appendices	n/a	Additional information, such as evidence of consultation with and feedback from the client and/or adviser.

Material to be submitted

A: Planning

Document describing the scenario, providing a rationale for the product, and defining success criteria

B: Solution overview

Design document outlining the solution

C: Development

Narrative of tools and techniques

D: Functionality and extensibility of product

Video demonstrating the functionality of the product

E: Evaluation

Commentary evaluating the product and recommending future improvements

Question	Avg. Mark	Highest Mark
Criterion A	3.4	6
Criterion B	3.0	6
Criterion C	5.8	12
Criterion D	2.3	4
Criterion E	2.9	6

Criterion A: Planning

Described an appropriate scenario for investigation for an identified client, providing evidence of consultation.

Provides a justified rationale for choosing the proposed product.

Provides a range of appropriate criteria for evaluating the success of the product.

6 marks.

Criterion A: Planning

Defining the problem

My client, Mrs xx, is a small business owner in xx. She sells sarees, an Indian traditional dress for women. Her transactions range from buying the raw materials for the sarees, sending them off to designers and actually selling the readymade sarees.

Presently, all of her business transactions are done manually. She keeps a stock list, a list of the raw materials she has sent off to designers (including the build of the sent items, their colour, the name of the designers and their relevant details), and a list of the readymade items for sale. She also uses the readymade list to calculate her remaining stock, items sold per month, and profit.

However, as her business expands and she has to deal with more and more transactions, such a method becomes inadequate. Moreover, since there is no efficient method of sorting items, she has faced considerable difficulty in tracking down lost materials in the past. Also, manual computation and management is becoming more and more tedious and eats up most of her time, affecting the productivity of the entire business.

I volunteered to take up the matter and write a GUI based software for handling all her transactions digitally, as well as develop a better sorting and management method for her materials. My CS teacher, Mr xx, agreed to be my advisor for this endeavour.

Word count: 228

Stating success criteria

i> An input which allows the user to add raw items (including build, and cost price).
ii> Creation of a database table that stores all the raw items added.
iii> A method of assigning unique id nos. to each raw item added.
iv> An output that allows the user to see the stock list sorted according to build.
v> An input that allows the users to enter the id numbers of sarees being sent off to designers, along with relevant information from those designers. The kind of design done for each saree and the price of the same also need to be recorded.
vi> A method of updating the raw items table with these information
vii> An output that allows the user to view a list of all items sent to designers with relevant details of the designers.
viii> An input that allows the user to add items returned from the designers.
ix> An input that allows the user to record any additional costs of the transactions.
x> Creation of a separate database table that stores all the readymade items.
xi> A method to move relevant items from the raw items list to the readymade items list.
xii> An output to view all the readymade items not yet sold.
xiii>A calculator that will allow the user to get the selling price of a particular saree. It should take profit (or profit percentage) and discount percentage as input.
xiv>If the saree is sold off, an input to indicate the same.
xv> An input that allows users to enter additional investments like salaries of employees, maintenance costs, etc.
xvi>A method to show the monthly and yearly investments or profits.

Rationale for proposed solution

I decided that choosing an OOP would be the best solution for Mrs. xx because of the complexity of the problem, and the amount of interactivity required between the graphical user interface and the functionality. I chose to use MS Access databases to store the data because it's the most widely used Office software program, and most other mainstream Office programs are designed to read MS-Access files.
After much consideration, I realized that creating an online system wouldn't be the best option because of the increased level of complexity it would lead to. Also, I couldn't guarantee enough security to make sure that no one could hack into an online. Plus, because her business was still restricted to a single retail store, it wouldn't be a big handicap. Java was chosen as the medium because it's platform independent, free and user-friendly. Moreover, writing a Java code meant flexibility to extend the product to android devices very easily, which could be required at some point.

Word count : 164

Sample 6

Criterion A: Planning

Computer Science for Kids

Defining the Problem

Client/advisor Ms xx is a primary school teacher at the European School; She wants to start teaching basics of Computer Science for elementary school children to give them basic knowledge and skills for future. She wants to start with covering several units in the beginning: how to represent numbers in computer, solve logical problems like finding missing terms in the numeric sequence, solving "Super Silly Squares" etc. It's not easy for her to find way which will be entertaining and educational and to get kids involved in such difficult subject.

In April, 2013 Ms xx came to our computer class during the brake and talked about her problem with our Computer Science teacher. She described what she wanted: to have computer program with friendly interface, because it would make children study with more joy.

As she was talking about her problem, I thought that it was good material for my Internal Assessment because Ms xx wanted to have IT solution for her problem. Computer Science teacher approved this idea.

In order to analyze this issue I decided to meet Ms xx to make an interview and find out details for my future software.

(xx, 2013)[1]

Rationale for Proposed Solution

I think effective java program helps me to solve Ms xx's problem. It's possible to create separate sections for teacher and for student. Teacher will be able to prepare activities for students in advance, in order to make children practice about certain topics.

For students, there will be made several game-type exercises according to unit they chose before starting game; Checking students works takes a long time and it requires hard work from teacher, thus program itself will be able to check whether student's answers. Java-program will "create" files (RandomAccessFiles) where teacher writes activities and reads data from it; thus, teacher can see what she has already used to avoid repetition also this will make her able print any material she wants.

According with these requirements, I decided to make my program in Java because of:

[1] xx (xx). Computer Science for Kids. (xx, Interviewer)

- I am learning it at school
- Platform independence
- With help of Netbeans IDE, process of writing program becomes more easy
- Using Java it's possible to provide user friendly interface for kids and for teachers
- With Graphical interface easy use buttons and assign them with different methods: delete, edit, add, clean, open window, check etc.
- After creating one type of activities for teacher and for student (methods, object, file), it will be easy to create another one (copy and change a little);
- Easy to manipulate with RandomAccessFiles: read and write in;

Stating Success Criteria

1. Program will allow teacher to add activities in txt file, change/edit, delete and print activities accordingly any time
2. Program will provide friendly interface for teacher and for kids
3. It will be easy for kids to follow instructions and solve problems/tasks generated by the program
4. Program will provide an option to enter any units/topic student/teacher wants
5. Program will provide a list of already used material to avoid repetition (for teacher)
6. Program will provide Auto Check of students work, throwing message weather answer is correct or incorrect
7. In case of data entry errors, program automatically gives warning messages to teacher
8. It will be easy to back up all files with activities created by teacher

Word Count: 427

Sample 8

Criterion A: Planning (6 marks)

The success criteria identified in criterion A will be used in criterion D to evaluate the effectiveness of the product.

Marks	Description
0	The response does not reach a standard described by the descriptors below.
1–2	An appropriate scenario for investigation for an identified client is stated. The rationale for choosing the proposed product is identified. The criteria for evaluating the success of the product are generally inappropriate.
3–4	An appropriate scenario for investigation for an identified client, providing evidence of consultation, is stated. The rationale for choosing the proposed product is partially explained and includes some appropriate criteria for evaluating the success of the product.
5–6	An appropriate scenario for investigation for an identified client, providing evidence of consultation, is described. The rationale for choosing the proposed product is justified and includes a range of appropriate criteria for evaluating the success of the product.

Criterion B: Solution overview

Provides a detailed and complete record of tasks
Provides a detailed and complete design overview including an outline test plan.

6 marks.

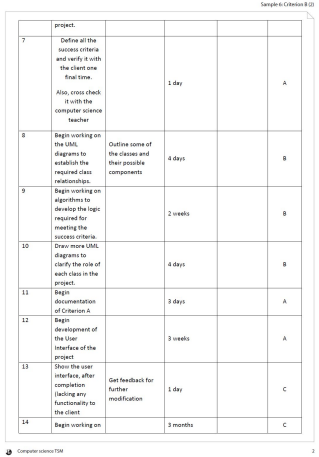
Criterion B: Record of tasks

Task number	Planned action	Planned outcome	Time estimated	Target completion date	Criterion

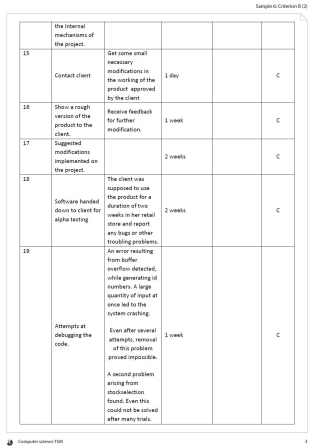
Record of tasks

Criterion B: Record of tasks

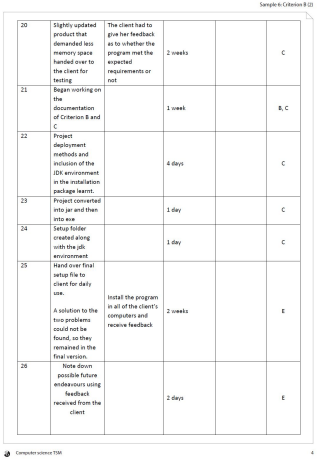
Task number	Planned action	Planned outcome	Time estimated	Target completion date	Criterion
1	Initial brainstorming of possible project ideas	A long discussion with my computer science teacher gave me insights on possible topics.	1 day		A
2	Contact client.	Discuss the possibility of digitizing her business procedures.	1 day		A
3	Discussion of the details of the project with client.	Discussion of the details of the project with client. Research into the scope of Java in solving these problems and list of the ideas that needed to be studied	1 week		A
4	Second discussion with client, further clarifying elements of the project.	Because intense research had been done on the scope of Java, some of the details needed to be modified, or cancelled altogether.	1 week		A
5	Learn some of the more developed gui concepts of Java.	The concepts not covered in the IB syllabus but necessary for the project had to be learnt.	1 month		A
6	Fix a schedule and development method for the		2 days		A



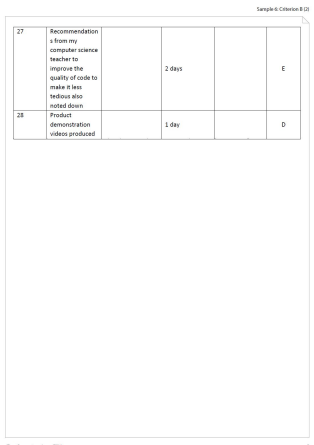
	project.				
7	Define all the success criteria and verify it with the client one final time. Also, cross check it with the computer science teacher		1 day		A
8	Begin working on the UML diagrams to establish the required class relationships.	Outline some of the classes and their possible components	4 days		B
9	Begin working on algorithms to develop the logic required for meeting the success criteria.		2 weeks		B
10	Drew more UML diagrams to clarify the role of each class in the project.		4 days		B
11	Begin documentation of Criterion A		3 days		A
12	Begin development of the User Interface of the project		3 weeks		A
13	Show the user interface, after completion (lacking any functionality to the client	Get feedback for further modification	1 day		C
14	Begin working on		3 months		C



	the internal mechanisms of the project.				
15	Contact client	Get some small necessary modifications in the sorting of the product approved by the client	1 day		C
16	Show a rough version of the product to the client.	Receive feedback for further modification.	1 week		C
17	Suggested modifications implemented on the project.		2 weeks		C
18	Software handed down to client for alpha testing	The client was supposed to use the product for a duration of two weeks in her retail store and report any bugs or other troubling problems.	2 weeks		C
19	Attempts at debugging the code.	An error resulting from buffer overflow detected, while generating id numbers. A large quantity of input at once led to the system crashing. Even after several attempts, removal of this problem proved impossible. A second problem arising from stackselection found. Even this could not be solved after many trials.	1 week		C



20	Slightly updated product that demanded less memory space handed over to the client for testing	The client had to give her feedback as to whether the program met the expected requirements or not	2 weeks		C
21	Began working on the documentation of Criterion B and C		1 week		B, C
22	Project deployment methods and inclusion of the JDK environment in the installation package learnt.		4 days		C
23	Project converted into jar and then into exe		1 day		C
24	Setup folder created along with the jdk environment		1 day		C
25	Hand over final setup file to client for daily use. A solution to the two problems could not be found, so they remained in the final version.	Install the program in all of the client's computers and receive feedback	2 weeks		E
26	Note down possible future endeavours using feedback received from the client		2 days		E



27	Recommendations from my computer science teacher to improve the quality of code to make it less tedious also noted down		2 days		E
28	Product demonstration videos produced		1 day		D

Record of tasks - Sample 6

Criterion B: Record of tasks

Task number	Planned action	Planned outcome	Time estimated	Target completion date	Criterion
1	Choosing a topic and discuss idea with future client	Chosen problem, willingness of Ms. xx to be client/adviser for my project and to collaborate with me	1 week		A
2	Discuss idea with computer science teacher	approved idea by teacher	5 days		A
3	First interview with Ms. xx	clear description of the problem	1 week		A
4	Tell computer science teacher my idea of problem solution for client	Java chosen to be used as a solution for client	3 days		A
5	Define criteria for success and finish with Criterion A	finished Criterion A; Criteria for success is defined	2 weeks		A
6	Start thinking on design of a product	Having an idea for the development of design and basic design is sketched up	1 week		B
7	create a schedule for developing the product	developed project development schedule in discussion with Ms xx	1 week		B
8	Work on the interface	Interface design is finished and agreed with Ms xx	1 week		B/C
9	Draw diagrams and flow charts in order to realize the program working process	Finished diagrams and process description flow charts	2 weeks		B
10	Think about how to test the product	Finished Detailed test plan	1 week		B

11	Start writing source code	Finished source code for main windows for teachers and students	1 week		C
12	Finding more information about Randomaccessfile methods	Researched information from the internet and Java books	1 week		C
13	Develop code and test every time to be ensure that everything works correctly	Different classes and methods are added in whole program; comments should be written that makes my code more readable; function testing should be made in every stage	4 weeks		C
14	Finish writing criterion C	Finished extended writing together with the source code	2 weeks		C
15	Test the final product	solution should be shown to Ms. xx and tested	1 week		D
16	Interview with Ms. xx	feedback from Ms. xx: testing results and suggestions about further developments	1 week		A/ E
17	Ideas how to improve the product	Written recommendations how to improve product in future	1week		A/D/E
18	Record video to show programs functionality	Video is recorded successfully	3 days		D

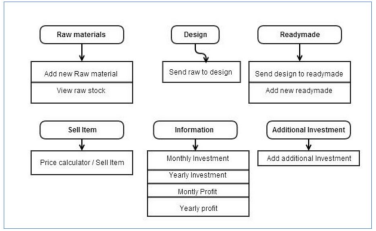
Record of tasks - Sample 8

Criterion B: Design

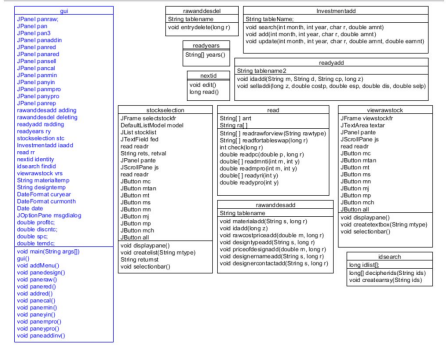
Elements required to be kept in the Menubar :

Raw	Design	Readymade	Sell	Information	Additional

Submenus for each menu:



Relationships between the classes



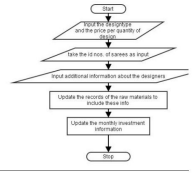


Figure 3 - Send raw to design flowchart

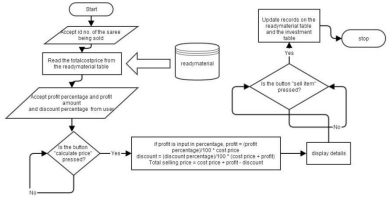


Figure 4 - Calculate selling price flowchart

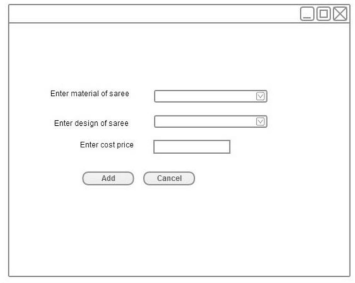


Figure 12 - Add new readymade material panel

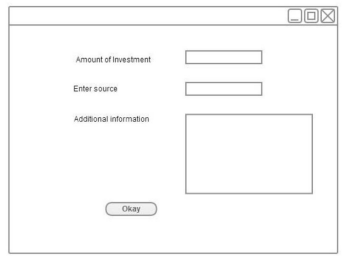


Figure 13 - Enter additional Investment panel

Design overview - Sample 6

Criterion B: Design

Activity A (Teacher's section)

Input data

Data	Example
Line 1 - 8 numbers of numeric sequence with 2 or more undefined members (defends on teacher) Line 2 – all 8 boxes filled with correct answer	Line 1 – 3 6 ? 12 ? 18 21 24 Line 2 – 3 6 9 12 15 18 21 24
Location - Text file (Possible to store maximum 30 record; record=task+answer)	

Output data

1. List of activities is displayed on the screen
2. List of activities is written in ActivityA.txt file (can be printed)

Activity B (Teacher's section)

Input data

Data	Example
Fill 25 squares so some boxes are empty and define sum of every row, column and diagonal; After clicking OK button - fill all undefined boxes with correct answer and click Add button.	Line 1 – 1 7 ? 12 23 Line 2 – ? 2 5 9 ? Line 3 – 7 ? 1 6 22 Line 4 – 6 4 2 ? 12 Line 5 - 14 ? ? 27 ? And 2nd five lines with correct answers
Location - Text file (Possible to store maximum 30 record; record=task+answer)	

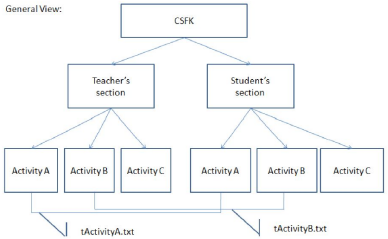
Output data

1. List of activities is displayed on the screen
2. List of activities is written in ActivityB.txt file (can be printed)

Activity C (teacher's section)

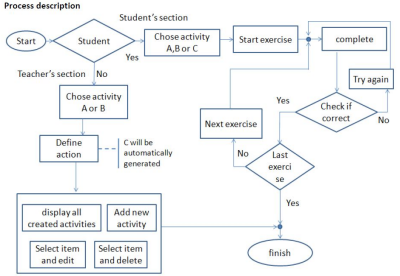
Input data - ! Everything is made by software– nothing will be saved

General View:



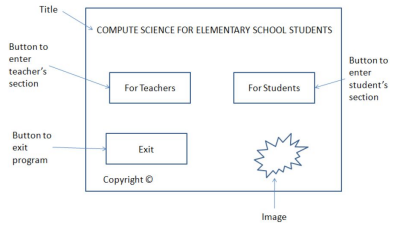
Activity C - the program creates tasks in students section

Process description



Interface

The main window



Schedule for developing the product

Program will be divided in three sections because of three different activities my client wants to create; each section contains two parts: "teacher" and "student". I think 2+2+1 weeks will be enough two complete all program; in section C I need to write code only for student's part.

Section 1 - Activity A for teachers (1 week)

- Create interface with buttons, textfields, labels, list objects
- Write code for add activity (data will be located in RandomAccessFile)
- Write code for Listview – teacher needs to see already created activities – data will be taken from file
- Write code for edit / delete item - edit=delete+add

Activity A for students (1 week)

- Create interface
- Write code to display first task
- Write code to check answer – answer will be compared with data written by teacher in txt file (task+answer are saved in txt file as one record)
- Write code to display next task (check if it's not the last one)

Section 2 - Activity B = 2 weeks (approximately the same)

Section 3 - Activity C = 1 week (I need to write code how to convert decimal in binary; check students answer and take next task will be the same as in Activity A and B)

Testing Plan

Examples, for testing the program:

Activity A

Teacher's Section – teacher should create tasks and answers

2	4	6	8	?	12	?	16

2	4	6	8	10	12	14	16

Max 30 exercises

Student's Section

2	4	6	8	?	12	?	16	condition
2	4	6	8	10	12	14	16	Student's answer

After checking each exercise, information massage will be displayed: "Correct! Please continue!" Or "Incorrect! Please try again".

Design overview - Sample 8

Test Plan

Action to be Tested	**Test method**
Unique no. being added to each new added raw material and readymade material	Input a large number of both new raw stock and readymade material and check the rawmaterial table and readymaterial table to see if all of them have been added with unique id nos.
Program is able to read and edit values in the database	Check the 'all' section in the viewrawstock Frame with the rawmaterial table to see if all the materials in the database are displayed. Update design information of some sarees using the sendrawtodesign panel and check to see if the changes have been reflected in the rawmaterials database
All the graphical interfaces are doing their appropriate functions	Run the program multiple to test individual features linked to graphical elements with various inputs and check if they are performing the desired functions.
Selection of sarees from a stock list is functional	While using the raw to design panel, input multiple ids by the select from stock option. Check the database for appropriate changes.
Check if the price calculator is functional	Check the readymaterial table for the raw data of the calculations, and manually check the results given with a calculator. If the values match, the calculator is functioning properly.
Check if the investment and profit reports are functional	Manually input data in the Financialchart table and then use the view investment and profit functions. If what was typed is returned, then the algorithm is functional.
additionalcosts being added to investments	Add additionalcosts multiple times both when sending design to readymade, and also using the add additional investment panel and check if the changes are being reflected in the Financialchart database.

Word count : 149

Test plan - Sample 6

Action Test	Method of testing & results
Test if the program runs and main window appears with two sections	Double click on the program icon; main window is displayed: teachers and students buttons works
Check if the teacher/student section works properly	Click on the buttons and appropriate windows appeared with three buttons and descriptions in each appropriately
Activity A and Activity B txt Files with title line should be crated automatically after clicking buttons first time	Files should appear in folder with program file; check if title line is written in it
In activity A (teachers section) check if the add function works properly; task and answer should be saved as one record/one line	Add new sequence with answers and check by refresh List viewer window; (it means ListView window works properly); also, check record in file
Each sequence should be deleted when pressed delete button	Deleted sequence should not appear in list after refresh and in file
Each activity should be edited	In list viewer should have different look after edit, also in file
Teacher should fill all fields	If fields are not field alert message should come and warn teacher to fill in everything
Student should fill all field	If fields are not field alert message should come and warn student to fill in everything
After finishing one exercise student will need to check and information massage should be displayed	Fill (do) one exercise and check on clicking button and appropriate message should be appeared on the screen; I will try both: correct and incorrect answers
Check weather all exercises will be shown on the screen or not when student plays	By clicking on the check button, if the answer is correct, next exercise will be displayed until the last one. If not – student should correct the answer and check it again.
Note: this testing plan will be used in all three section: Activity A, Activity B and Activity C In Activity C student can take as many activities as she/he wants	

Words = 112

Test plan - Sample 8

Criterion B: Solution overview (6 marks)

- The student must provide a record of tasks and a design overview that includes an outline test plan.
- The *Record of tasks* form must be used.
- The record of tasks and design overview must refer to the product proposed in criterion A.

Marks	Description
0	The response does not reach a standard described by the descriptors below.
1–2	The record of tasks and the design overview, including an outline test plan, are limited. From this information it is difficult to see how the product was developed.
3–4	The record of tasks and the design overview, including an outline test plan, are partially complete. They provide a basic understanding of how the product was developed.
5–6	The record of tasks and the design overview, including an outline test plan, are detailed and complete. From this information it is clear how the product was developed.

Criterion C: Solution overview

Identify techniques used in developing the product.
Explain the techniques, with screenshots, that were used to develop the product.
Explain why techniques have been used and why they are adequate for the task.

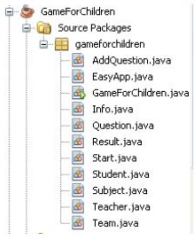
12 marks.

The mysterious ‘development document’

- The student must present a list of the techniques used in developing the product.
- The techniques may include algorithmic thinking, data structures, software tools and user interface. This list need not be exhaustive but should illustrate how the major components of the product were developed.
- The student must provide evidence of algorithmic thinking. <- systematic approach to solving problems
- The information in the development documentation must provide a detailed account, using extended writing and other appropriate information, to explain the following:
    - The structure of the product and why it is appropriate
    - The algorithmic thinking used in the development of the product
    - The techniques used in the development of the product and reasons why they are appropriate to it (may include screenshots, exemplar data, reference to information in the appendix)
    - Any existing tools that are used in the development of the product, such as code libraries, software packages, web hosting, security information or infrastructure issues
- Any reference material such as templates, program code, applets or other materials that have been used or modified must be acknowledged in this criterion. The code used in the product can be included in the appendix.

Criterion C: Development

All classes:



List of the techniques:

1. Each class extends EasyApp[1] to create controls and place them in the window;
2. Different methods are used in the program: methods that are defined in EasyApp.java and methods created by me;
3. String array
4. RandomAccessFiles: Math.txt, Art.txt, Music.txt, SC.txt, UOI.txt, team.txt, Results.txt
5. Static methods and variables

Program file:



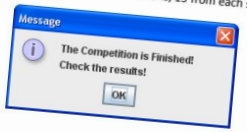
[1]Mulkey, Dave. "Free and Easy Java." 2004-2007. http://ibcomp.fis.edu/easyJava/ (accessed xx).


```
static String point;  // how many point gain the team
void check() {   //is the answer correct or incorrect; if yes, how many point the team gained
    String answer = Answer.getText();
    if (answer == null || !answer.equals("A") && !answer.equals("B") && !answer.equals("C")
            && !answer.equals("D")) {
        outputString("Enter Correct Answer!\nTry again...");
    } else {
        if (answer.equals(Start.questionArray[i][6])) {
            point = Start.questionArray[i][7];
        } else {
            point = "0";
        }
        outputString("You gained " + point + " point");
```

After checking the answer – seqond teams turn is to choose question of the particular subject, value and answer it. I used following way to distinguish between teams and add correct values.

```
if (count % 2 == 0) {
    Start.team2array[count / 2 - 1][0] = subject;
    Start.team2array[count / 2 - 1][1] = val;
    Start.team2array[count / 2 - 1][2] = point;
} else {
    Start.team1array[count / 2][0] = subject;
    Start.team1array[count / 2][1] = val;
    Start.team1array[count / 2][2] = point;
}
```

After answering 30 questions, 15 from each side game will automatically finish by the message:




```
void Question() {
    subject = Subject.getSelectedItem();  // subject the question belongs to
    value = Value.getSelectedItem();  // question value 1,2,3
    count++;
    if (count <= 30) {  //max questions = 30 - 15 for each teams
        // 15 = 5 subject * 3 questions from each subject
        if (count % 2 == 0) {  // even number question - for team II
            teamName.setText(team2);
            teamsName = team2;
            new Question();
        } else {
            teamName.setText(team1);  // odd number question - for team I
            teamsName = team1;
            new Question();
        }
    } else {
        outputString("The Competition is Finished!\nCheck the results!");
    }
```


Narrative - Sample 7

Criterion C: Development

Techniques used :

Libraries Imported

```
//technique- importing packages
import java.sql.*;
import java.util.Date;
import java.text.*;
import java.text.DateFormat;
import java.text.SimpleDateFormat;
import java.util.Calendar;
import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
```

SQL(Structured Query language) is used for the program to communicate with the MS-Access databa
to hold the data.

util.Date, util.DateFormat, text.SimpleDateFormat and util.Calender are used for collecting and form
the dates of input.

awt and swing are used in the graphical components.

SQL commands used

```
addRow= "INSERT INTO "+tableName+" (month, year, investmentonraw) VALUES("+ month + " , " + year +" ,"+
```

Figure 1 - adding rows

```
updateRow= "UPDATE "+tableName+" SET investmentonraw="+ (amnt+examt) +" WHERE month="+ month +" AND yea
```

Figure 2 - Updating rows

```
String tableName= "RawandDesign";
ResultSet rs=stmt1.executeQuery("Select id,gonefordesign from "+tableName);
rs= stmt1.getResultSet();
```

Figure 3 - Reading fields from a table

```
String updateRow= "DELETE FROM "+tablename+" WHERE id="+ r +"";
stmt1.execute(updateRow);
```

Figure 4 - Deleting rows

Production of the GUI

The gui class handled almost all of the GUIs except for viewing the raw stock, and selecting
materials from a list.

Using JFrames and JPanels


```
ButtonGroup bgdes= new ButtonGroup();

bgdes.add(batik);
bgdes.add(embroidery);
bgdes.add(acid);
bgdes.add(aplick);
bgdes.add(hand);
bgdes.add(block);
```

Buttongroups were used where only one selection from a list of Radiobuttons was necessary.

Getting the present month and year and thereby updating the Financialchart table

```
public DateFormat curyear = new SimpleDateFormat("yyyy");
public DateFormat curmonth = new SimpleDateFormat("MM");
public Date date = new Date();
```

Figure 5 - Global Declarations

```
// updating the financialchart table
int cury= Integer.parseInt(curyear.format(date));
int curm =Integer.parseInt(curmonth.format(date));
lsadd.search(curm, cury, 'd', (Double.parseDouble(designprice.getText())*desarr.l
```

Figure 6 - Calling the search method from the Investmentadd class

The search method in the Investmentadd class takes the current month, year, an inves
identifier, and the amount invested as parameters.

The investment or profit identifier tells the function which column to edit the infor
investments on raw materials are denoted by an 'r'.

Similarly, investment on design – 'd'

Additional investment – 'a'

Profit –'p'

It then checks whether the month and year are present in the table. If it is, t
a new row with the information.


```
public void paneaddinv()throws Exception
{
  paneadd.setLayout(null);
  JTextField amr= new JTextField();
  final JLabel aml= new JLabel("Amount of Investment");
  JTextField src= new JTextField();
  JLabel sll= new JLabel("Source");
  JButton okie = new JButton("Okay");
  JTextArea addin= new JTextArea();
  JLabel addinlb= new JLabel("Additional information");
  JScrollPane jsqa= new JScrollPane(addin);
```

Figure 89 - Adding the Graphical Components (Source: Appendix – 1- iii)

```
okie. addActionListener(new ActionListener()
  public void actionPerformed(ActionEvent e)
  {
    try {
      int cury= Integer.parseInt(curyear.format(date));
      int curm =Integer.parseInt(curmonth.format(date));
      lsadd.search(curm, cury, 'p', Double.parseDouble(aml.getText()));
    }catch(Exception z){}
});
```

Figure 90 - Adding the additional investment to the FinancialChart table

Selecting id numbers from stock list

A number of thinks needed to be kept in mind while designing the stockselection list, and yet at the end, it did not seem to work properly, for reasons I couldn't understand.

The first problem was to decide how to generate a dynamic list system that allows for multiple selections. I had decided to create a linked list and add JCheckBox-es for selection. However, extracting only the identity numbers from the selection became too much of a hasle.

JList seemed to be a pretty good alternative at the early phases when I wasn't aware of the problems. It allowed for multiple selections intrinsically, and extracting strings from selected components were easier.

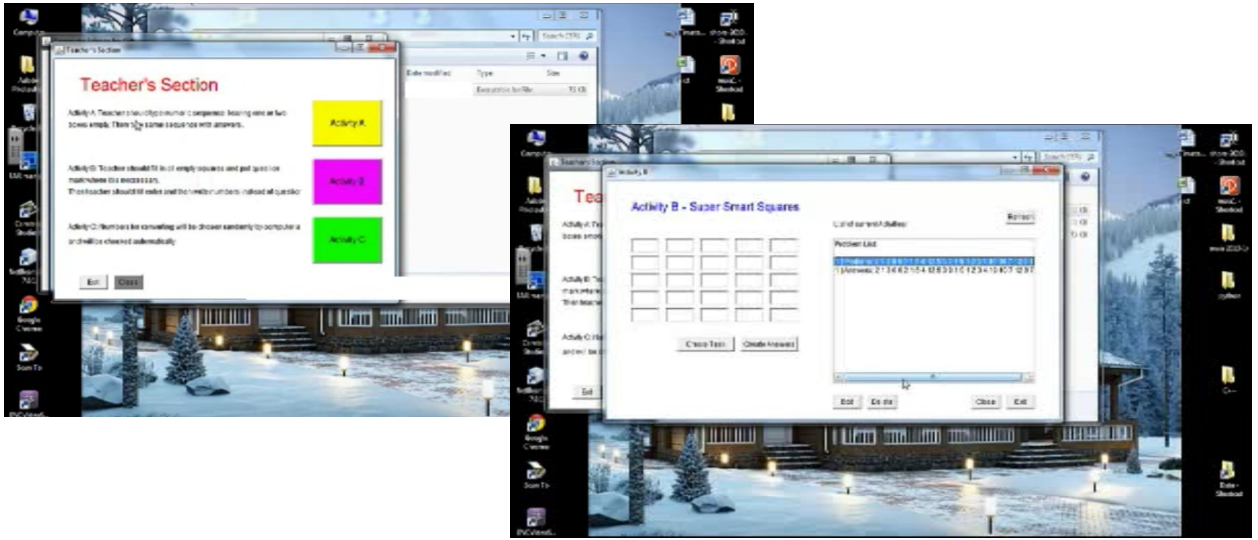
Narrative - Sample 7

Marks	Description
0	The response does not reach a standard described by the descriptors below.
1–4	The use of techniques demonstrates a low level of complexity and ingenuity or does not address the scenario identified in criterion A. It is characterized by limited use of existing tools. There is no explanation of why the techniques are used or how they are adequate for the task. Sources are used but are not identified.
5–8	The use of techniques demonstrates a moderate level of complexity and ingenuity in addressing the scenario identified in criterion A. It is characterized by some appropriate use of existing tools. There is some attempt to explain the techniques used and why they are adequate for the task. All sources are identified.
9–12	The use of techniques demonstrates a high level of complexity and ingenuity in addressing the scenario identified in criterion A. It is characterized by the appropriate use of existing tools. The techniques are adequate for the task and their use is explained. All sources are identified.

Criterion D: Functionality and Extensibility of Product

Demonstrate that the product functions well.
Code enables straightforward expansion and modifications of the product.

4 marks.



Video of product, with voice over (about 5 mins)

Criterion D: Functionality and extensibility of product (4 marks)

This criterion assesses the extent to which the product:

- functions, as evidenced in the video
- can be expanded and modified by future users as evidenced in the design and development documentation.

Marks	Description
0	The response does not reach a standard described by the descriptors below.
1–2	The video shows that the product functions partially. Some expansion and modification of the product is possible but difficult.
3–4	The video shows that the product functions well. Some expansion and modifications of the product are straightforward.

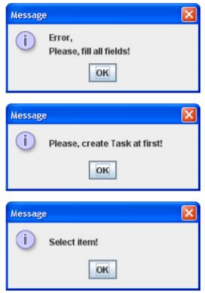
Criterion E: Evaluation

Evaluate the product fully, against the success criteria identified.
Include feedback from the client/adviser.
Provide realistic recommendations for further improvement of the product.

6 marks.

Criterion E: Evaluation

Success Criteria

Teacher's ability to add activities in txt file, change/edit, delete and print activities any time	Activity A, Activity B in teacher's section has create, edit and delete functions; they work correctly. Every created activity is saved in corresponding file, which can be printed.
Friendly interface for teacher and for kids	Interface was approved by client during the development and testing stages.
It will be easy for kids to follow instructions and solve problems/tasks generated by the program	While testing program on kids, there was no trouble for them to follow instructions; Students solved problems, created by the teacher
An option to enter any units/topic student/teacher wants	Because of three sections, there is an option to enter and chose any topic.
A list of already created tasks to avoid repetition (for teacher).	In teacher's section list of already created activities will be displayed on screen
Auto Check of students work	Check button in student's section checks answers providing corresponding message
Warning messages to teacher in case of error.	Was tested and worked successfully every time. 
It will be easy to back up all files with activities created by teacher	It's very easy to copy and save already created files with tasks in order to create new ones.

Recommendations for Further Development

Password for teacher's section – adding login and password for client, will prevent students from seeing answers.

Despite the fact that in teacher's section teacher can view all created activities, list can get too big and it will be uncomfortable for teacher to search entire list to make sure activity is not repeated. Problem can be solved by generating code to compare already created activities with newly created one (search method in array).

Program has no abnormal data and it works in any case; it will be better to check all entered data whether they are numbers or not using parseInt method.

Program checks student's answer but does not count the amount of correct/incorrect answers. By making something like "hall of fame" students will be able to see who and when got the most amounts of correct answers (all students will need enter names and save information in txt file).

Add more different logical activities – it will be easy to write the same delete, edit and createarray codes for different ones.

Word Count = 376

Short document

Criterion E: Evaluation (6 marks)

- The student must evaluate the effectiveness of the product based on feedback from the client/adviser. This must include direct references to the success criteria identified in criterion A.
- The student must recommend proposals for the future improvement of the product.

Marks	Description
0	The response does not reach a standard described by the descriptors below.
1–2	There is a limited attempt to evaluate the product against the success criteria identified in criterion A. There is limited evidence of feedback from the client/adviser and any recommendations for further improvement are trivial or unrealistic.
3–4	The product is partially evaluated against the success criteria identified in criterion A including feedback from the client/adviser. Recommendations for further improvement of the product are largely realistic.
5–6	The product is fully evaluated against the success criteria identified in criterion A including feedback from the client/adviser. Recommendations for further improvement of the product are realistic.